

User Interface (UI) and User eXperience (UX) design

Software Engineering meets
Human-Computer Interaction

Software Engineering Course – 08

2025-2026

The plan for today

Today, we will:

- Talk about **what design is** and why it's so **important**

- Discuss the **main types of design** relevant to **software development**

- Learn how to perform **user research**

- Sketch **wireframes** to model our user's path through our app

- Create interactive **prototypes** for rapid **experimentation**

- Validate** our ideas

- Perform **usability testing**

- Ensure our app is **accessible** to all

What is design?

And why do we need it?

What is design?

How would **you** define it?

What is design?

design (verb)

to create, fashion, execute, or construct according to **plan**

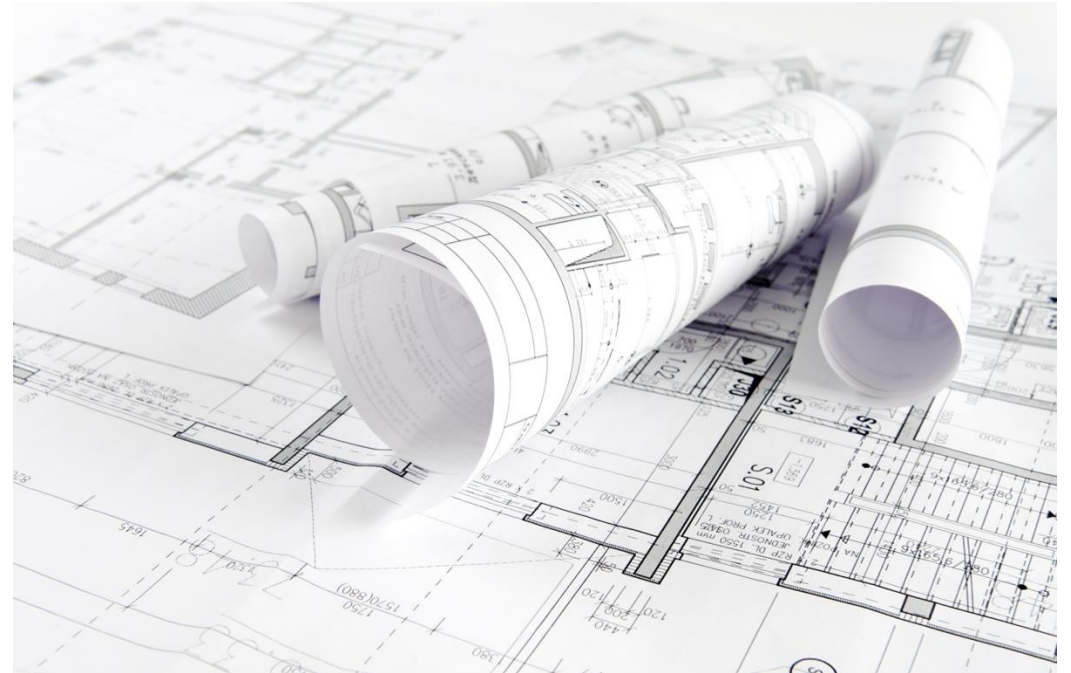
to **conceive** and **plan** out in the mind

to **devise** for a **specific function** or **end**

design (noun)

a particular **purpose** or **intention** held in view by an individual or group

deliberate purposive **planning**



According to [the Merriam-Webster dictionary](#)

Why do we need design?

Being deliberate about the **purpose** of our work and the **decisions** we make helps ensure we develop **the right product**, which helps our users to **achieve** their **goals** more easily.



Examples of bad design



"Norman" doors

Your **product** = Lots of small **decisions**

(+ actually implementing them)

Design → Smarter decisions

Importance of design

A badly designed interface can lead to:

- Frustrated** users

- Low** productivity

- Increased risk of **mistakes**

All of these can make the business **lose customers!**

Benefits of good design

A well-designed product can lead to:

Happy users who use your product **more**

Increased **brand recognition** and **customer loyalty**

Increased productivity

⇒ users complete tasks **faster**

Reduced risk of mistakes during operation

No need to re-design/update product as much after it is released

⇒ **reduced** long-term maintenance **costs**

Product is **usable** by a **wider range of users** (accessibility)



“By failing to prepare, you prepare to fail.”

Benjamin Franklin

By failing to design, you design to fail.

Types of design

What kind of designer are you?

Types of design

There are many branches/sub-disciplines of design:

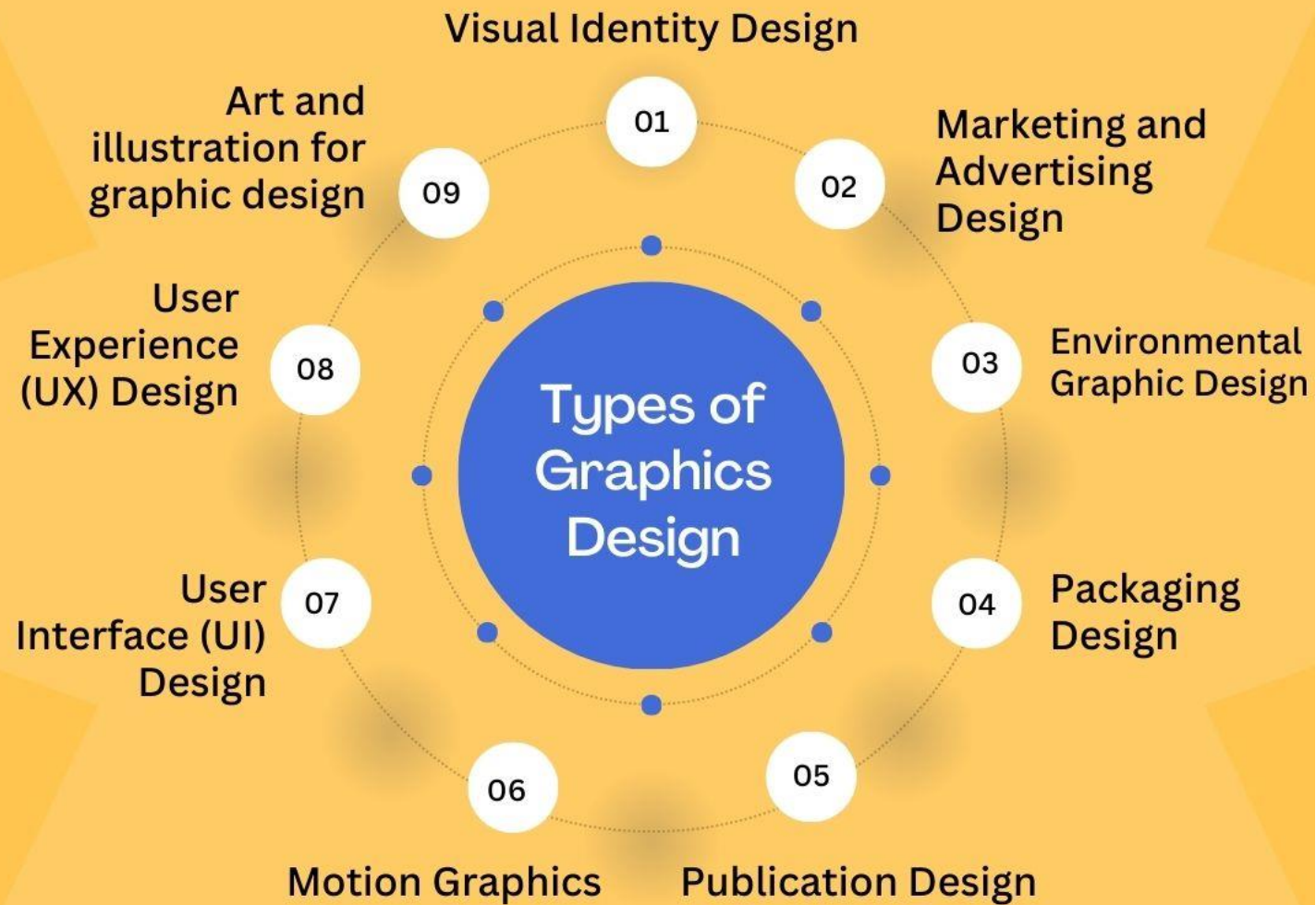
- Graphic design
- Sound design
- Communications design
- Industrial design
- Web design
- etc.

As software engineers, we're only interested in a subset of these!

Types of design

The kinds of designing which we will be interested in are:

- **Software design** – what this and other courses are about, in general (includes software architecture)
- **Graphic design** – includes a lot of subdisciplines which are relevant for software development
 - User interface design
 - Motion graphic design
 - ...
- **User experience design** (which goes beyond thinking just about user interfaces!)
- (Video) **game design**
- ...



Let's go through
a few of the **main branches of design**
to better understand them!

Canva



UI design

Explore colour
palettes

Liam



Browse



Where to today?



EXPLORE SYDNEY



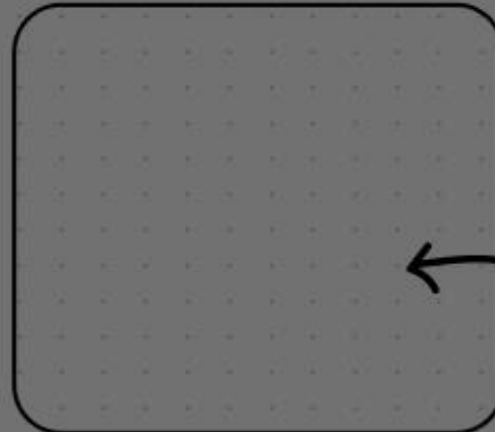
EXI

Pheobe

BOOK TODAY



Locations



Sydney

Lorem ipsum dolor sit amet consectetur. Tortor urna sed massa eget leo imperdiet elit sit nunc. Eu sit bibendum congue elit vitae leo lectus justo quam. Malesuada parturient habitasse nec

Darcy

User interface design

The user interface refers to any sort of **interface** – boundary or medium – through which a prospective user **interacts** with your app or product.

UI designers are specialists which focus on making the interfaces **easy to use** and **pleasant**.

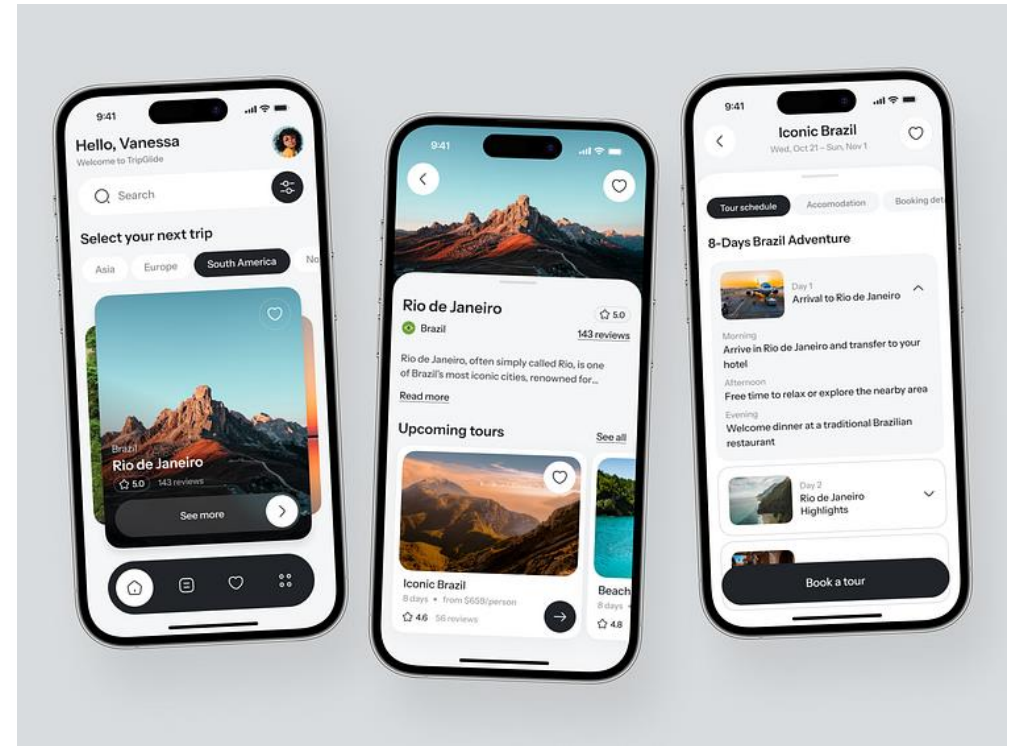


Image found on [Dribbble](https://dribbble.com),
a popular platform for designers

Human-computer interaction

Human-computer interaction (HCI) refers to the process through which human users engage with and operate computer systems.

Usually, HCI refers to the study of all kinds of **interactions/interfaces** between humans and devices:

- Visual** (graphical) interfaces (GUI)

- Auditory** feedback (notification sounds, speech recognition)

- Tactile/haptic** feedback (think touch screens and phone vibrations)

Did you ever develop an app with a **graphical** user interface?

What about an app with an **auditory** feedback? (e.g. make a bell sound)

How about **haptic** feedback? (e.g. smartphone vibration)

User experience design

User Experience

(noun)

Design

(verb)

Image source

User experience design

User experience (UX) design refers to the **holistic** design approach where you consider not only what the user **sees** or **interacts with** (user interface), but also:

- what their **needs** are

- how your product makes them **feel**

- what kinds of **experiences** it offers them

User experience design

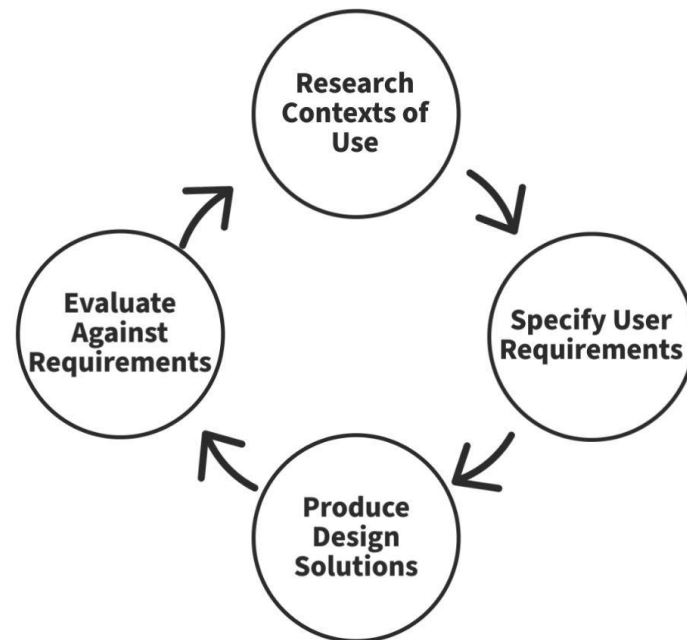
“No product is an island. A product is more than the product. It is a **cohesive, integrated** set of **experiences**. Think through all of the stages of a product or service – from **initial intentions** through **final reflections**, from the first usage to help, service, and maintenance. **Make them all work together seamlessly.**”



Don Norman, American professor and designer who coined the term “**user experience design**”

User experience design

UX design is part of the larger shift towards **user-centered design**, an iterative design process which involves real users at almost every step of the way.



User-centered design

Typical **mindset** of software developers/programmers is **system/technology-centric**: how the code works (modules, libraries, database etc.)

User-centered mindset: focus on how humans use the code (screens, interactions, flows, feedback etc.)

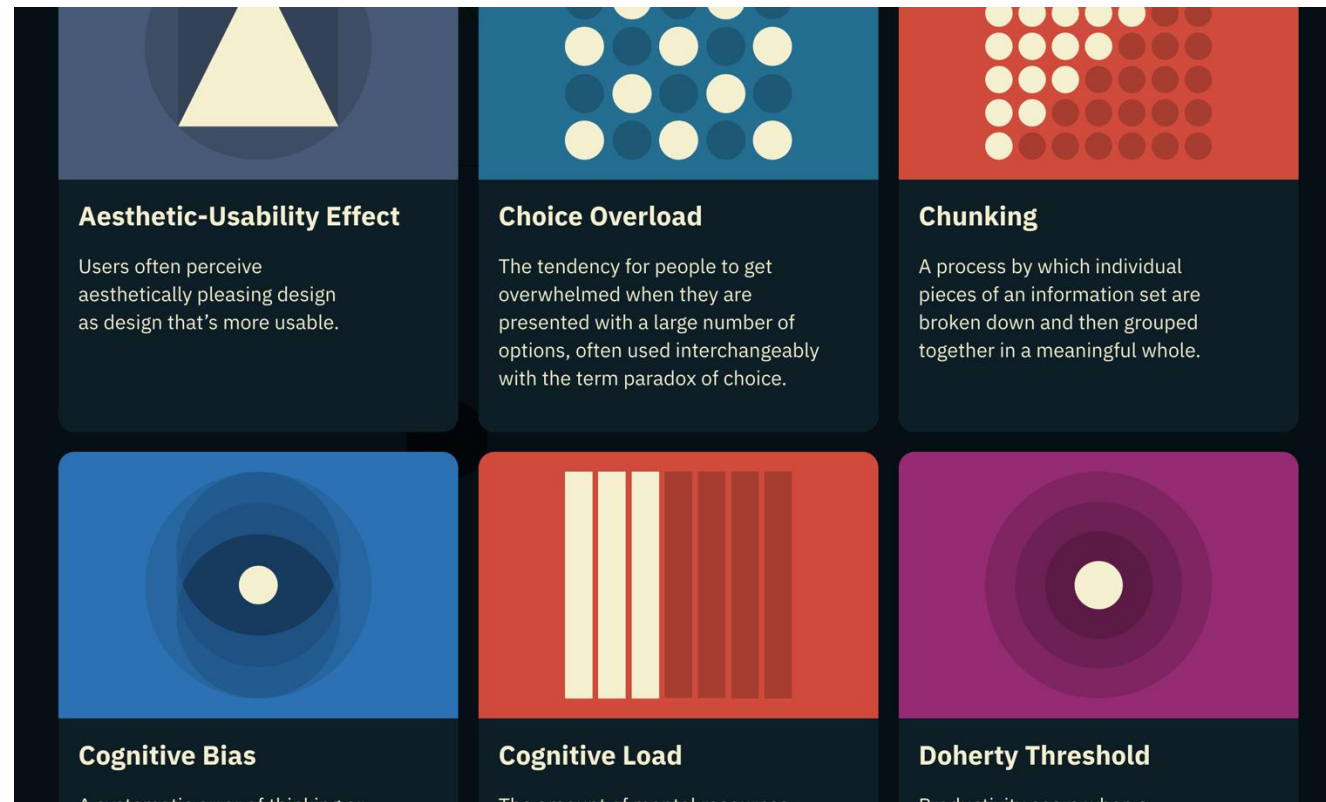
UI vs. UX design

The main **difference** between UI design and UX design is that

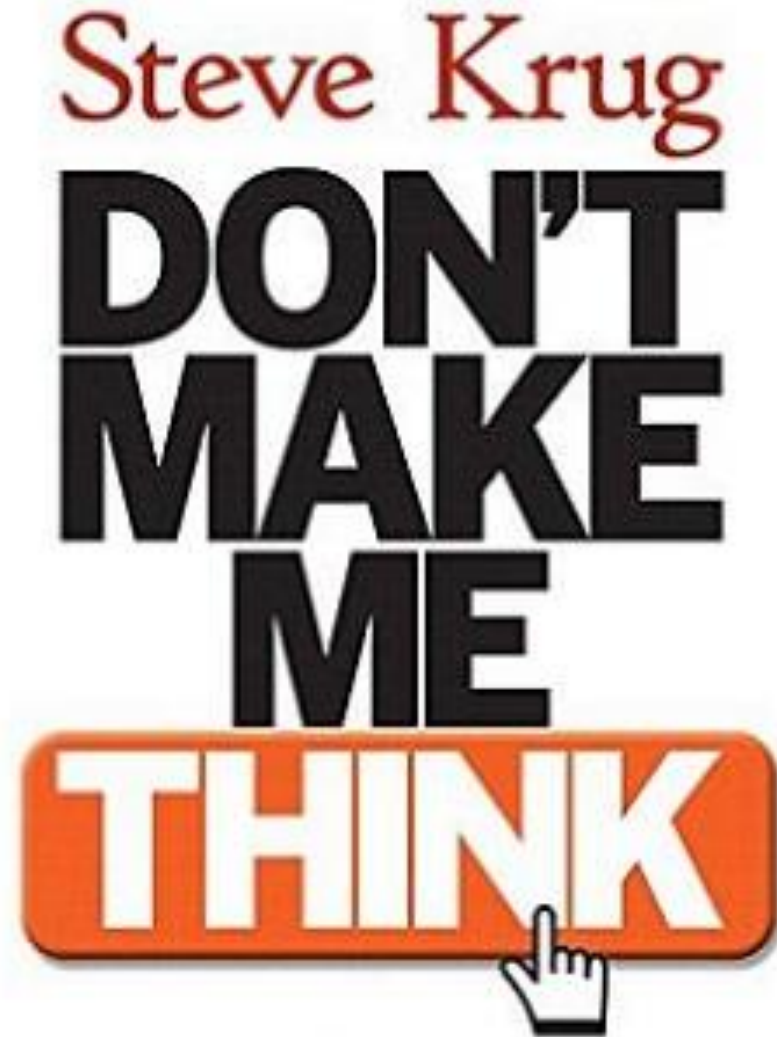
UI design focuses exclusively on how to design a better human-computer **interface**

UX design studies all aspects of a user's subjective **experience** with the app: usability, usefulness, desirability, satisfaction, overall performance etc.

Laws of UX



[Laws of UX](#) website



A Common Sense Approach to Web Usability
FOREWORD BY ROGER BLACK

Don't Make Me Think!

A recommended book for those who design and develop web/mobile apps.

[Book link](#)

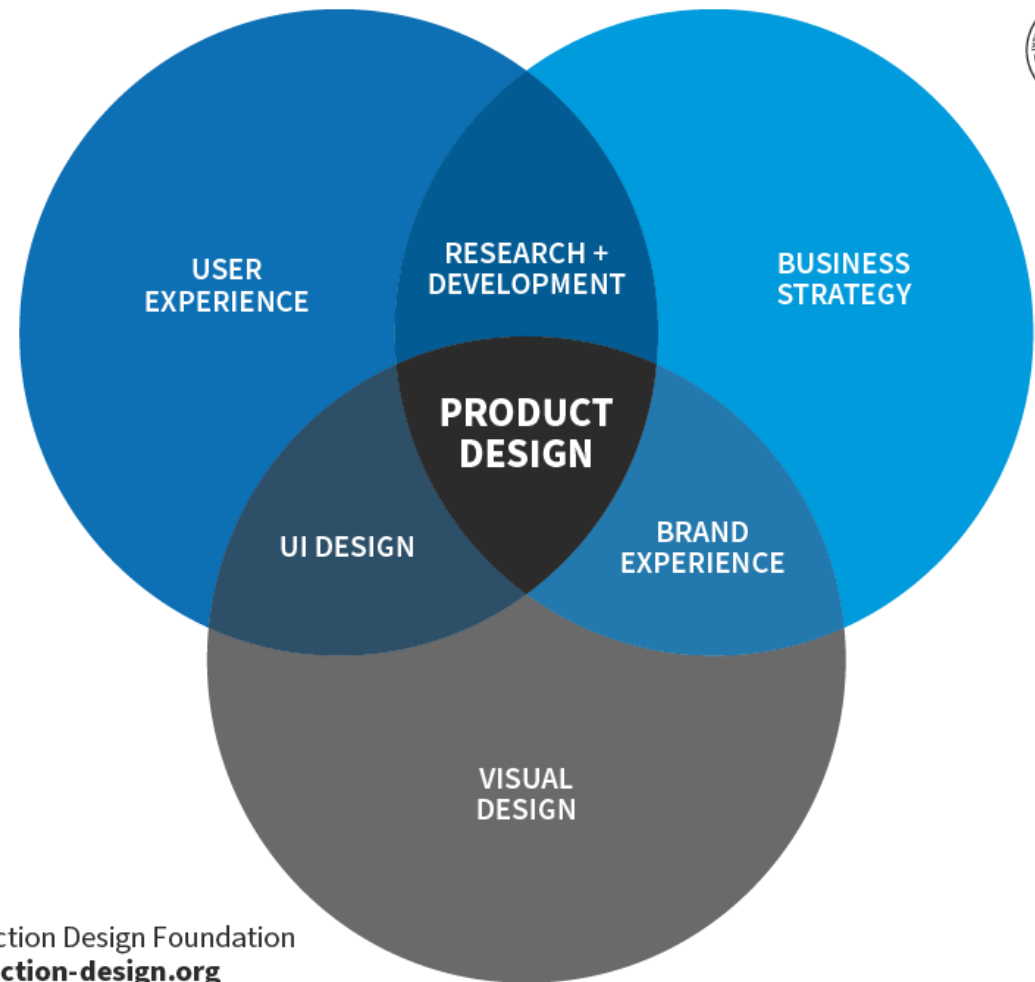


Product design

Product design

This is the creative process through which **new products** (or **services**) are **developed**, which the business then sells to customers.

Your **product** (app) should help the **user** in a certain **context** to complete a **task**, which will help them achieve their **goal**.



The message is clear: **to design better (software) products, we need to understand our users better.**

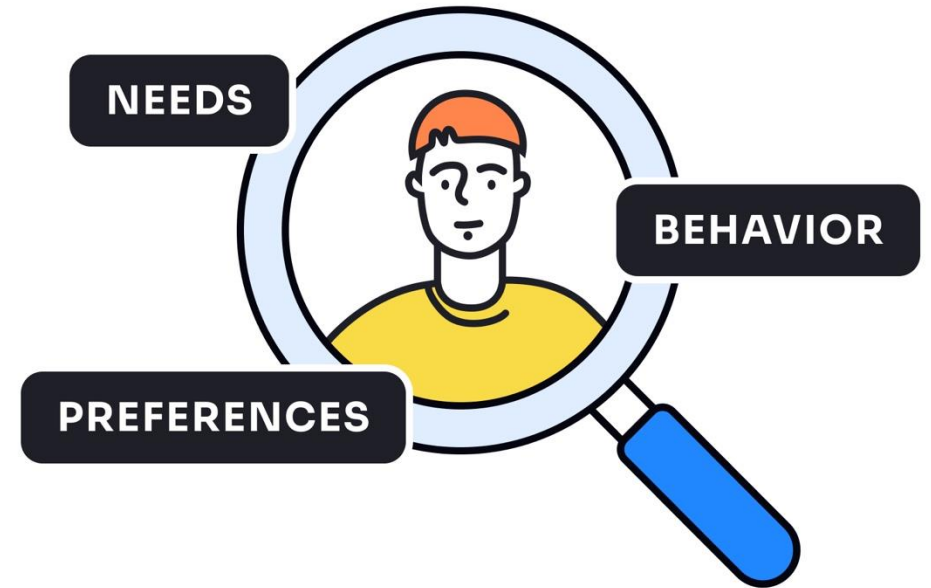
How do we do so?

User research

User research

User research involves **understanding** existing or potential users, their **needs, behaviors, desires, abilities/skills, context** etc.

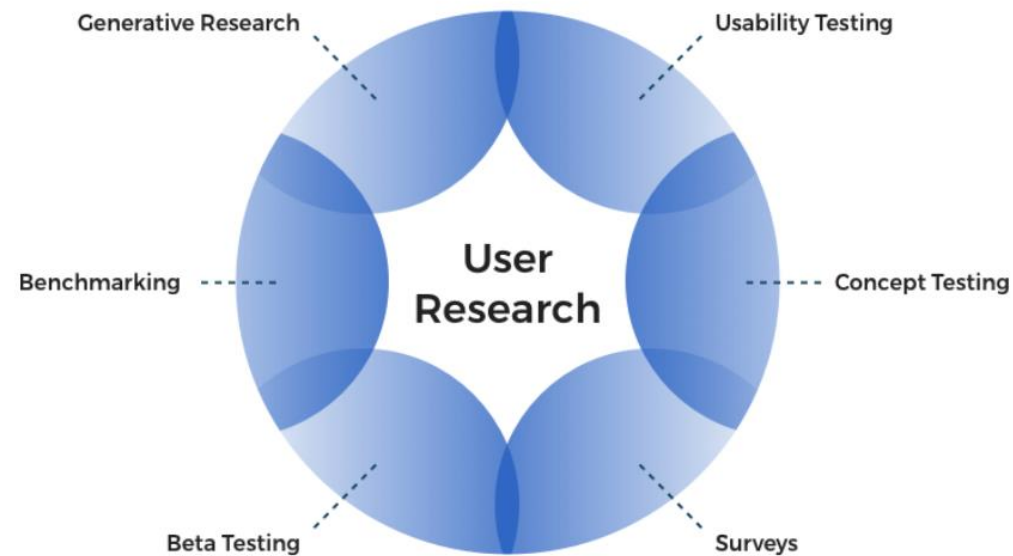
It is an exercise in **empathy**; you want to understand well who they are and how they use your product. The research methods we use are inspired by the ones used in sociology and psychology.



User research methods

User research methods

There are a plethora of approaches for performing user research, mostly inspired by existing sociological study methods.



[Image source](#)

User interviews

The simplest way to find out what your users **think about your app** is to ask them directly.



User interviews

User interviewing usually involves **finding** people from the right group, then inviting them over for a shorter or longer **discussion** about their needs in relation to your product/app. Nowadays, a lot of such interviews are conducted **online**, over videocall platforms.

However, there is one big **risk** / problem associated with user interviews: they might **not be** very **informative**, or **users might tell you what you want to hear** (especially if you use **leading questions**!)

"Ounce for ounce, there's no better way to learn what customers want and will buy than this wonderful little book. If you want your new product or new business to succeed, start here."

— John Mullins
Author of *The New Business Road Test*



Rob Fitzpatrick

THE MOM TEST

How to talk to customers and learn
if your business is a good idea
when *everyone is lying to you*

[Recommended book](#)

Intercept survey method

Similar to the interview method, but you talk with users while they are “in their natural environment”: at their workplace or in another location relevant for your business domain.

Example: you want to develop an app for **book-lovers** to track their reads, so you go to **the local library** and ask random people if they’d be willing to talk with you for 5 minutes. Their feedback is much more relevant than that of people from the coffee shop.



Survey method

One of the easiest ways to obtain quantitative feedback on your app/design idea is to use a **survey / form**, a predefined set of questions which you ask many people to fill in.

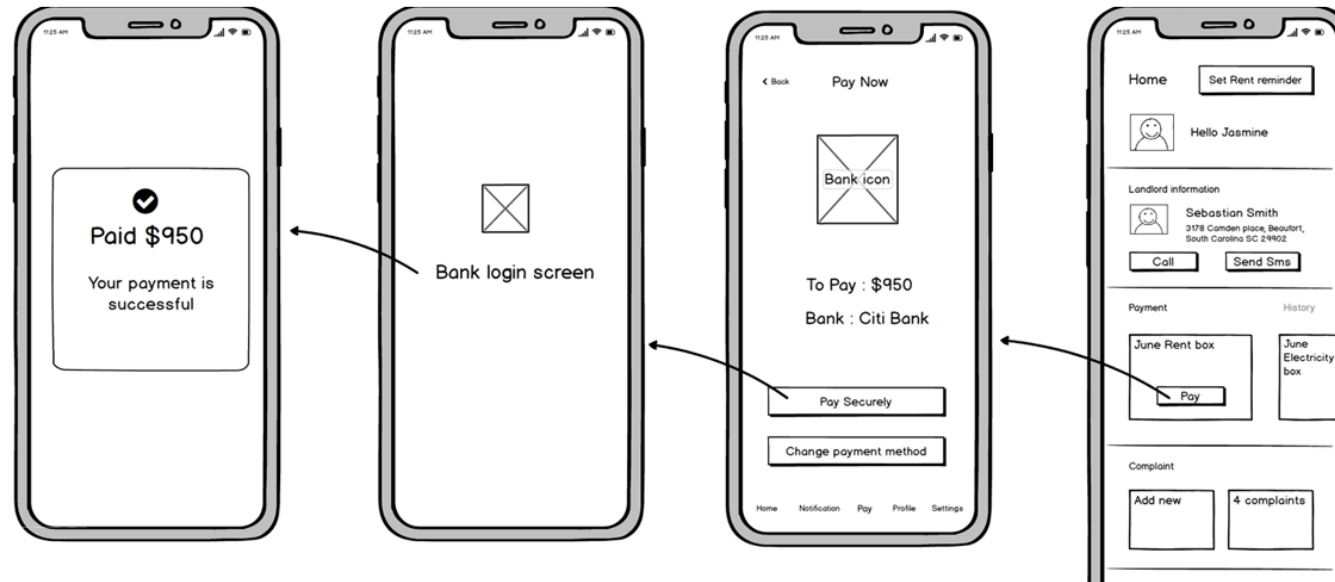
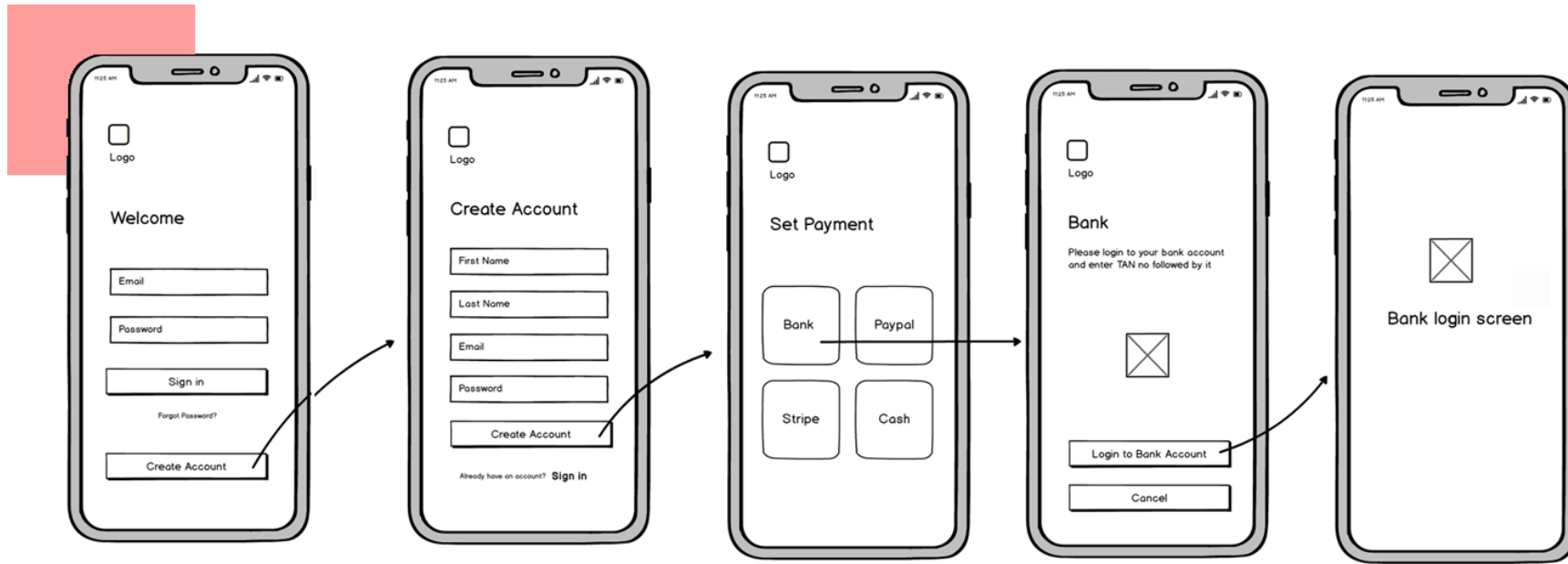
Thinking about your most recent purchase experience at **Amazon.com**, how satisfied are you with...

	Not at all satisfied	Slightly satisfied	Moderately satisfied	Very satisfied	Completely satisfied
...how often you are notified when something is discounted, on sale, or on promotion	☐	☐	☐	☐	☐
...the ability to determine when a product is discounted/on promotion	☐	☐	☐	☐	☐
...the ability to filter down to just items that are discounted, on sale, or on promotion	☐	☐	☐	☐	☐
...your ability to know when you are getting the best deal	☐	☐	☐	☐	☐
...features that notify you when a product is discounted, on sale, or on promotion	☐	☐	☐	☐	☐
	Not at all satisfied	Slightly satisfied	Moderately satisfied	Very satisfied	Completely satisfied
...the number of deals/promotions you usually find	☐	☐	☐	☐	☐
...the ease of understanding how long a deal/promotion will last	☐	☐	☐	☐	☐
...how frequently you're able to find deals/promotions	☐	☐	☐	☐	☐
...the ability to know whether you are getting the best value on a product	☐	☐	☐	☐	☐
...features that help you understand how much you are saving on a product	☐	☐	☐	☐	☐

Wireframing

Wireframing

Once you have gathered enough information about your users and have some idea for what the app should do, it's now time to start designing a low-fidelity non-interactive **wireframe** for your app.



[Image source](#)

Wireframing

You should focus on the aspects most relevant to the users' experience: the **screens** they will see and interact with, the **navigation**, the **menus**, etc.

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components with
when you upgrade to
plan.

Keyboards



Typography



Buttons



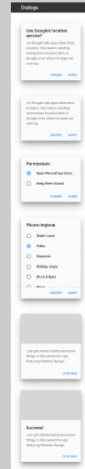
Toolbars



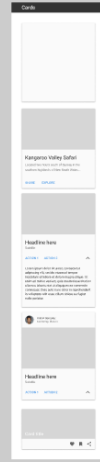
Bottom...



Dialogs



Cards



Lists



Text Fields



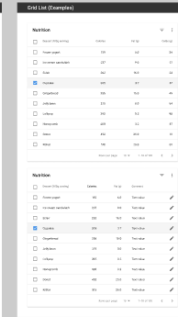
Sliders



Grid lists



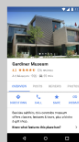
Grid lists - Exa...



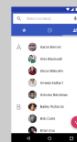
Wireframing tool: Figma

Examples

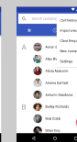
Places



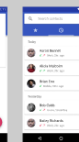
Phon...



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List ...



Form...



Form...



Card...



Card...



Exte...



Musi...



Photos



Phot...



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Design

Prototype

Page

CCCCCC

100

Variables

Styles

Text styles

Ag Display 4 · 112/112

Ag Display 3 · 56/56

Ag Display 2 · 45/48

Ag Display 1 · 34/40

Ag Headline · 24/32

Ag Title · 20/32

Ag Subheading (Mobile) · 16/24

Ag Subheading (Desktop) · 15/24

Ag Body 2 (Mobile) · 14/24

Ag Body 2 (Desktop) · 13/24

Ag Body 1 (Mobile) · 14/20

Ag Body 1 (Desktop) · 13/20

Ag Caption · 12/20

Ag Button · 14/24

Color styles

Primary App Color

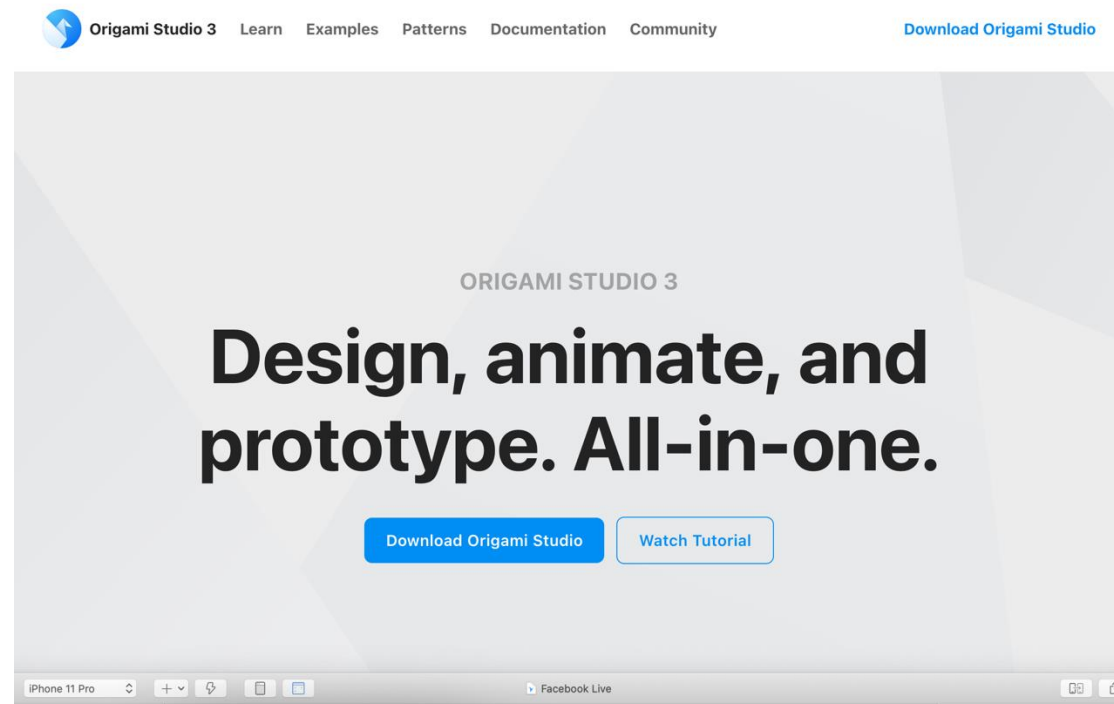
Prototyping

Prototyping

Once you have the wireframe sketched out and you believe you understand what your users' needs are and how your product can satisfy them, it is time to **validate / test** your hypothesis.

In theory, you could go ahead and develop a fully fleshed-out product and proceed to test it with real users, but at this stage it's cheaper and smarter to instead develop a low-fidelity **prototype** which you can show to your users.

Prototyping tools – Origami Studio



The prototyping tool developed and used by Meta (e.g. for Facebook): origami.design

Usability testing

Usability testing

While you might find the app/product to be very easy to use, this is probably not the case for other people who are unfamiliar with it!

Usability testing helps us gather **feedback** and **metrics** from real users, to understand what are the **potential pain points** for them when trying to use our app.

It is a kind of a **manual testing** method which is done as part of a product's **quality assurance** cycle.

Usability Testing: Flow of Information



Accessibility

One app to be used by all!

Accessibility

Accessibility means designing your app/product to be **usable** by all people, especially those with **disabilities** (for example, visual impairment, locomotor disability etc.)



Advantages of accessibility

It makes your app usable by the **largest possible number of people** – larger **potential user/customer base**

It helps **regular users** be **more productive** with the app

It can help you comply with **laws** and **regulations**

European accessibility act

The [European accessibility act](#) is a directive that aims to improve the functioning of the internal market for accessible products and services, by removing barriers created by divergent rules in Member States.

≡ Contents

European accessibility act

Products and services covered

Preparatory documents

Related news

Related events

European accessibility act

The [European accessibility act](#) is a directive that aims to improve the functioning of the internal market for accessible products and services, by removing barriers created by divergent rules in Member States.

Businesses will benefit from:

- common rules on accessibility in the EU leading to costs reduction
- easier cross-border trading
- more market opportunities for their accessible products and services

Accessibility guidelines



[W3C Accessibility Guidelines \(WCAG\)](#)

[Understanding the Web Content Accessibility Guidelines](#)

Questions?

Disclaimer: I am not a professional UX designer

Kahoot!

Thank you
for your participation!